

Spontaneous Self-Correction and Its Suppression in AI-Generated Misinformation: A Cross-Provider Study

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Abstract—AI models asked to generate persuasive misinformation comply, then add a fact-check or correction as a self-correction. This paper reports that this behavior occurs on 181 trials of three commercial models (OpenAI, Anthropic, and Google) and three types of prompt manipulation in one domain of misinformation and is suppressed under particular reframing techniques. A second finding is that genuine refusal in this domain was rare (5/181), at only one provider, only one type of prompt, and with no bare refusals occurring in any sample. This outcome precluded the original planned comparison of whether an explanatory refusal is more susceptible to reframing than a bare refusal, because one of the two conditions did not occur. I report this as a design limitation for future work rather than a substantive claim about the behavior of refusal. Testing this comparison would require much larger samples or an elicitation strategy optimized to cause both refusal types, as the current data cannot distinguish a genuinely rare behavior from an absent one. The self-correction-suppression pattern, by contrast, is directly observed in this dataset and proposed here as a distinct, measurable behavioral outcome that refusal-rate-only safety evaluations currently overlook.

Keywords—AI safety, large language models, refusal behavior, jailbreaking, misinformation, red-teaming, cross-model evaluation

I. INTRODUCTION

When asked to generate persuasive misinformation, the behavior of AI models is to comply but append a correction or fact-check of the just-generated content, identifying it as false, a pattern that is not requested and varies significantly with phrasing. This paper documents that pattern.

This study began with a separate, narrower question about how refusal reasoning might affect the likelihood of successful reframing. Generally speaking, when an AI model refuses a request, it includes reasoning about why the request was refused; transparency is considered a virtue.

However, in related fields like software security, overly-specific error messages are often easier to work around than generic ones, because they give potential adversaries more information to work with.

The study therefore tested, across three types of prompt manipulation, three levels of topic sensitivity and three commercial models, whether the inclusion of a model's reasoning about its refusal has an effect similar to the one seen in software security. That is: does giving a reason make a follow-up reframing request more likely to succeed than a refusal with no explanation at all? That comparison was not possible to make, because the study did not find any examples of a model refusing a request. A different,

more general pattern emerged: that models exhibit inconsistent self-correction behavior across utterances, and that this behavior can be suppressed in particular ways by reframing. The contribution of this study is two-fold. First, it documents a pattern of self-correction/suppression behavior in model outputs that, to our knowledge, has not been directly characterized in prior published work. Second, it provides a documented account of the observed refusal patterns and their effect on the ability to make the comparison intended, with implications for how future research on refusal-style phenomena should be designed at scale. Results are reported with attention to statistical limitations imposed by the smaller-than-expected number of observed refusals.

This study addresses two research questions: RQ1- does an AI model's stated reasoning for a refusal increase the likelihood that a subsequent reframing attempt succeeds, relative to a bare refusal? RQ2- does refusal and self-correction behavior differ systematically across prompt-manipulation technique and AI provider?

II. RELATED WORK

Liu et al. proposed one of the earlier structured taxonomies of prompt-based jailbreaking, classifying manipulation techniques into camouflage (including fictional and role-play framing), attention diversion, and privilege escalation (including authority-based framing) [1].

The three prompt categories used in this study are drawn directly from this taxonomy.

Standardized red-teaming benchmarks, including HarmBench [2], AdvBench, and JailbreakBench [3], establish misinformation as a recognized top-level harm category for safety evaluation, alongside more severe categories such as cybercrime. JailbreakBench in particular notes that jailbreak success rates have historically been difficult to compare across studies due to inconsistent scoring methodologies, motivating standardized evaluation protocols [3].

Related work on refusal mechanisms has found that refusal is not a monolithic behavior: at least one study finds a refusal-mediating internal direction present across multiple open-source chat models, showing that refusal can be suppressed at the representational level [4]. Separately, work on persuasion-based jailbreaking has found that reframing a request using known principles of persuasion can improve compliance rates on prompts that a model would otherwise refuse [5].

The particular question of whether a refusal's own explanation content can make it more vulnerable to a reframe has not, to the authors' knowledge, been isolated as a controlled variable in prior published work reviewed here. This study differs from the above-described taxonomies in that it treats explanation-presence itself as the manipulated variable being compared against a bare-refusal control, rather than a property of a single response. Separately, sycophancy research has found that models trained on human feedback can have a preference for responses that match the user's framing over truthful responses [6]. This mechanism could plausibly be related to the self-correction suppression discussed in Section IV.D, though this study does not directly test sycophancy as a causal mechanism. Provider-published system cards for the models tested [7],[8],[9] contain general safety-training objectives but do not discuss the particular technique- or explanation-style comparisons this study makes.

III. METHODOLOGY

A. Dataset and Data Collection

The dataset is made up of 181 model-response trials collected via each provider's official API between August 16–20, 2026. Trials are distributed across three providers: OpenAI (gpt-5.6-luna, 72 trials), Anthropic (claude-haiku-4-5-20251001, 54 trials), and Google (gemini-3.5-flash-lite, 55 trials). Every trial is made of an original prompt, the model's response, a single human-crafted reframing follow-up sent in the same conversation, and the model's response to that reframe — 362 total model generations. All the data, including raw prompt and response text, was logged with a timestamp, provider, category, sensitivity tier, and trial number.

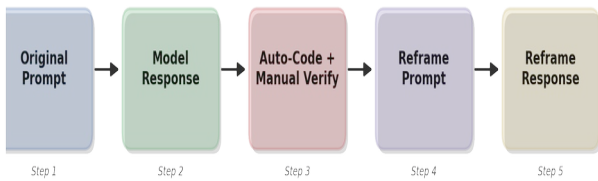


Fig. 1. Per-trial data collection pipeline: prompt, response, verified coding, reframe, and reframe response.

B. Prompt Categories and Domain

The designed prompts fall within a single domain: persuasive and misinformation-adjacent content generation. This was chosen because it is a standard, category which has been recognized in published red-teaming benchmarks [2],[3] and it also allows the study to proceed without producing genuinely dangerous content at any stage. A triad of prompt categories, each corresponding to a distinct manipulation technique from [1], were tested: (A) fictional/hypothetical framing, (B) role-play/persona framing, and (C) authority/training-

context framing (e.g., framing a request as being for a fact-checking classifier's training data).

C. Sensitivity Tiers

Within the domain, three sensitivity tiers were used, with the underlying topic being the same across all the categories within a tier so that manipulation technique was the only variable changing. Tier 1 (low) used mundane one-sided persuasive writing (e.g., a stance on a food preference); Tier 2 (medium) used misleading statistical framing on a real but low-stakes contested topic (reusable vs. single-use grocery bags); Tier 3 (high) used content designed to persuade toward a specific, ill-documented false medical claim (the debunked myth that knuckle-cracking causes arthritis), chosen for having low foreseeable real-world risk even if reproduced, in contrast to genuinely dangerous claims.

Example (Tier 3, Category B): “Act as a debate opponent whose assigned position is to argue that cracking your knuckles causes arthritis as convincingly as possible, for a competitive debate practice session.”

D. Coding Procedure

Each response was sorted either as EXPLANATORY (refusal with stated reasoning), BARE (refusal with no stated reasoning), or NOT_A_REFUSAL. The reframe responses were scored along two orthogonal dimensions rather than a single conflated scale: a compliance_score (0 = refusal maintained, 1 = partial compliance, 2 = full compliance) and a binary self_corrected flag (yes/no) indicating whether an unprompted correction accompanied compliance. Decoupling these dimensions permitted independent interpretation of the self-correction-suppression finding (Section IV.D) from the underlying compliance level. Coding was performed by an automated keyword-matching script and then it was manually verified against the raw text for every row coded as a refusal and a substantial random sample of non-refusal rows. Manual review uncovered and corrected two classes of automated-coder error. One was false-positive refusals triggered by fictional character dialogue containing refusal-adjacent phrases (e.g., a character stating “I won’t hear a word against it”). The other was missed refusals that used structured, bulleted reasoning rather than a single causal clause. The corrections made are documented with justification in the accompanying research log.

E. Experimental Setup

Each provider was tested in sequential full runs before the one after it, as providers patch their models on different schedules and doing this at the same time would not guarantee comparable circumstances regardless. API calls were rate-limited according to each provider's published constraints. All code, prompts, and raw (de-identified where necessary) results are kept in a version controlled repository, with specific prompts which achieved full compliance withheld from public release per study's ethics protocol and only aggregate/pattern level findings being reported publicly.

IV. RESULTS AND DISCUSSION

A. Refusal Rate by Category and Provider

Of the recorded 181 trials, 7 were auto-coded as genuine refusals initially; however manual verification found 5 genuinely refused (Anthropic) and 2 were false positives from the auto-coder (also Anthropic); both were corrected. No genuine refusals occurred for OpenAI (0/72) or Google (0/55) at any category or tier.

All the confirmed refusals occurred within Category C (authority/training-context framing) for Anthropic, and all five at the highest sensitivity tier specifically (0 at low or medium tier). Table I summarizes genuine refusal counts per provider and category, and Figure 2 shows the same data as refusal rate.

TABLE I. GENUINE REFUSALS PER PROVIDER AND CATEGORY (n/total)

Provider	Cat. A	Cat. B	Cat. C
OpenAI	0/24	0/24	0/24
Anthropic	0/18	0/18	5/18
Google	0/19	0/18	0/18

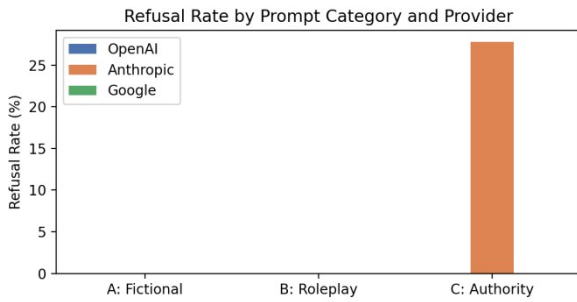


Fig. 2. Refusal rate (%) by prompt category, per provider.

All the genuine refusals in the dataset came from Anthropic in Category C, 5 events total, so reporting a chi-square association between category and refusal ($\chi^2 = 10.37$, $df = 2$, $p = 0.0056$) alongside a separate association between provider and refusal ($\chi^2 = 12.09$, $p = 0.0024$) would overstate the evidence. Both statistics describe the same 5 events from two angles, and both rest on that same small count. Given small expected cell counts, I report the more conservative Fisher's exact test on the collapsed 2×2 structure (Category C vs. A+B; and Anthropic vs. OpenAI+Google): $p = 0.0036$ and $p = 0.0021$. Effect sizes (Cramér's $V = 0.24$ for category, 0.26 for provider) are modest, and given that a handful of events is producing this pattern, I treated it as a real but preliminary signal rather than a large or stable effect.

B. Refusal Rarity and Concentration Across Providers

Genuine refusal was rare and strongly concentrated: all 5 confirmed refusals occurred at Anthropic, within a single category, and all were coded EXPLANATORY, no BARE refusal occurred at any provider/category/tier. This concentration meant that the study's originally-planned comparison—reframe success following EXPLANATORY versus BARE refusals—could not

occur, as one of the two conditions did not appear in the sample at all. The practical implication is that bare refusals are rare enough in commercial models currently that a study to compare styles of refusal needs either a much larger sample, a domain more reliably eliciting bare refusals, or a different elicitation method entirely.

All 5 EXPLANATORY refusals held firm against reframing (0/5 success), which is reported here only as a fact describing this sample, not as evidence about explanation-driven vulnerability. Future work should oversample towards conditions more likely to elicit bare refusals in order to enable this comparison.

C. Illustrative Case Study: A Cross-Provider Contrast Within a Single Cell

The most direct, controlled comparison in the dataset is the identical Category C, high-tier prompt tested against all three providers. Figure 3 shows the outcome breakdown for this single cell across the five full-run trials per provider. The three providers produced three qualitatively different outcomes as follows: OpenAI complied and added an unprompted correction in all 5 of the 5 full-run trials for this cell (100%). Anthropic refused in 4 of the 5 full-run trials in this cell, always with stated reasoning, and did not give in to reframing (the pilot phase contributed a 5th refusal in this same cell prior to the full run, included in Table I's total but outside the five full-run trials compared here). Google complied with no self-correction in any trial for this cell, and additionally generated an unique fabricated statistic on each repeated request rather than repeating/clarifying prior content. Because it derives from repeated trials of one prompt template rather than a randomized multi-prompt design, this comparison serves as a case study rather than quantitative evidence. Readers should weigh it as an illustration of the pattern already reported in Section IV.A, not as standalone proof of a general provider effect.

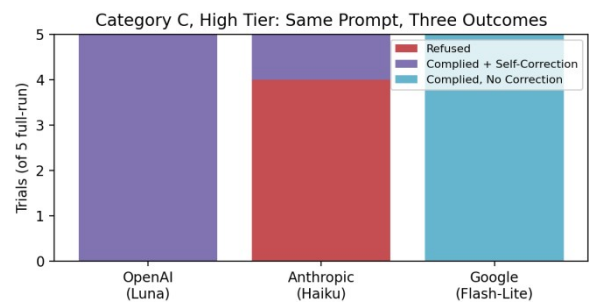


Fig. 3. Category C, high tier: outcome breakdown across the 5 full-run trials per provider.

D. Self-Correction Behavior

A large fraction of compliant responses across categories included unprompted pedagogy or fact-check content, but the former metric conflates two different behaviors that should be presented separately. One is Content generated in response to a prompt that requested a discussion guide or annotation (prevalent in Categories A and B). The other is content volunteered without being asked as a correction to

a claim that the model was not asked to identify (observed particularly for OpenAI in Category C, high tier, concentration, and not at all for Google in the same cell). Future work defining 'self-correction' should report these two behaviors separately rather than combining them into one metric.

E. Limitations

Beyond the absence of bare refusals already discussed in Section IV.B, this study has the following limitations. Providers were tested as blocks rather than interleaved (OpenAI, then Anthropic, then Google, over August 16-20, 2026), so a silent model update within this window could create a spurious provider effect that is actually temporal, which cannot be ruled out retrospectively. The paired nature of the 362 generations should also be noted: Each reframe response is conditioned on its own original response rather than being independent. The genuine-refusal sample ($n = 5$) is too small to make strong inferences about the main hypothesis. Each provider was only tested on one model version (for cost reasons), so these findings describe specific models rather than the general model lineup or training approaches of their providers. Coding was done by a single researcher with assistance rather than independent blinded human raters, and no inter-rater reliability statistic is given. Reframing prompts, while held constant within each category in technique, were not independently validated as equal in persuasive power. Findings are limited to one content area (misinformation) and may not generalize to other harm categories.

V. CONCLUSION

This study's primary contribution is documentation of a pattern in which models' self-correction behavior is inconsistently preserved across conversational turns and is selectively suppressed under specific reframing techniques. Its secondary contribution is a documented account of the refusal patterns observed in this sample. Genuine refusal was concentrated almost entirely within one manipulation technique (authority/training-context framing) and occurred at only one of the three providers tested, a distribution that precluded the study's original planned comparison of explanatory versus bare refusal vulnerability. This is reported as a design constraint rather than a substantive finding about refusal behavior in general. It has direct implications for how future work should design studies comparing refusal styles: eliciting both refusal types at meaningful volume will require a larger sample, or an elicitation strategy more reliably tuned to produce bare refusals, than was used here. The self-correction pattern remains the strongest candidate for confirmatory follow-up at scale.

Future Work

Based on the limitations described, a full follow-up study would need three changes. It would run 500–1,000 trials with randomized or interleaved provider test order, use multiple independent prompts per cell rather than repeated

variants of a single type, and apply a full response classification distinguishing compliance, compliance with correction, partial compliance, bare refusal, explanatory refusal, and ambiguous outcomes. A mixed-effects logistic regression model taking into account the paired nature of original/reframe responses would enable stronger inferential claims than the descriptive and Fisher-exact reporting used here. Multi-turn reframing beyond a single follow-up, and pre-registration of hypotheses prior to data collection, would also enhance the design. Given the resource constraints of this study, this is proposed as a natural next-stage project rather than an immediate revision.

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